Vellore Institute of Technology
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Digital Systems Design (BECE102L)

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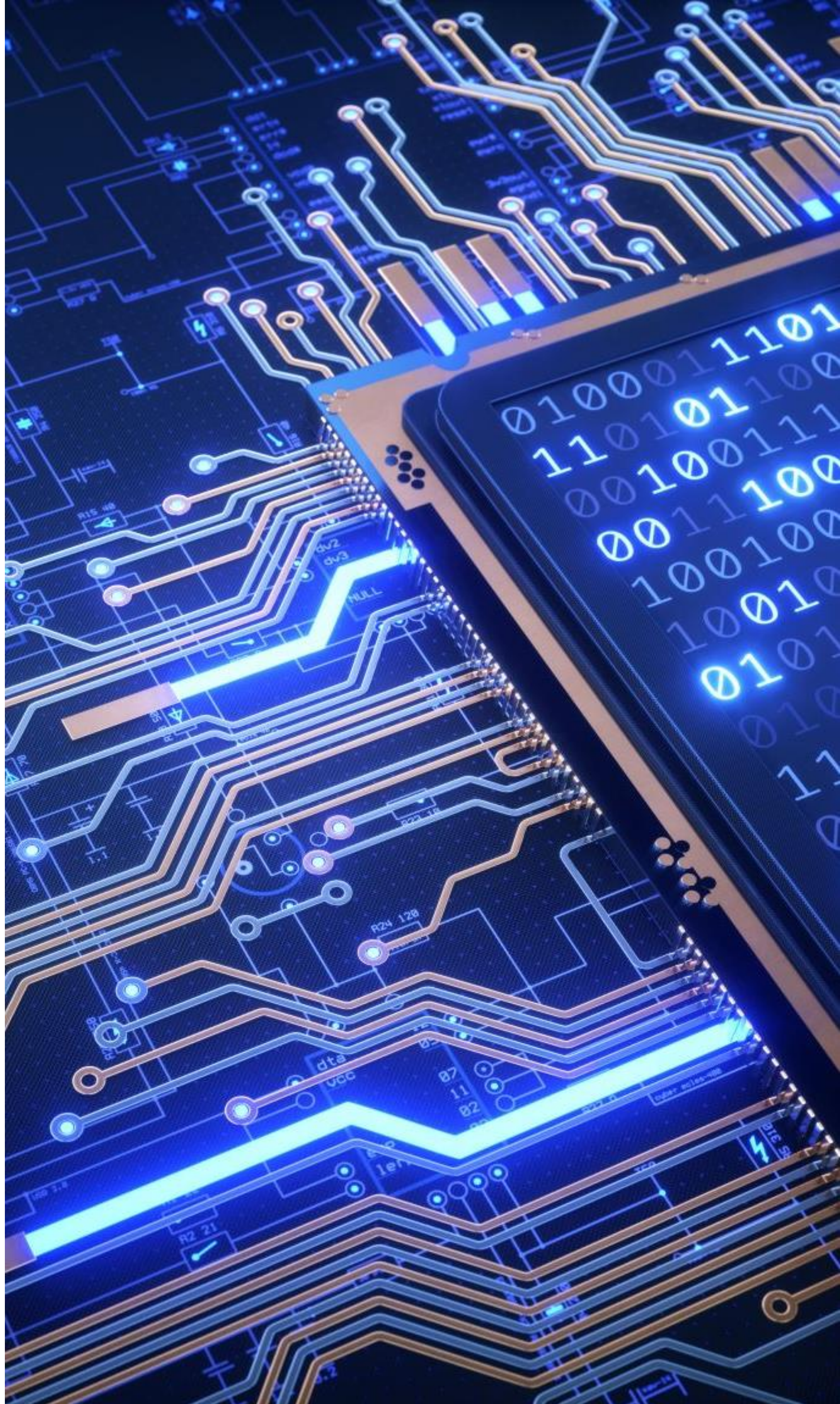


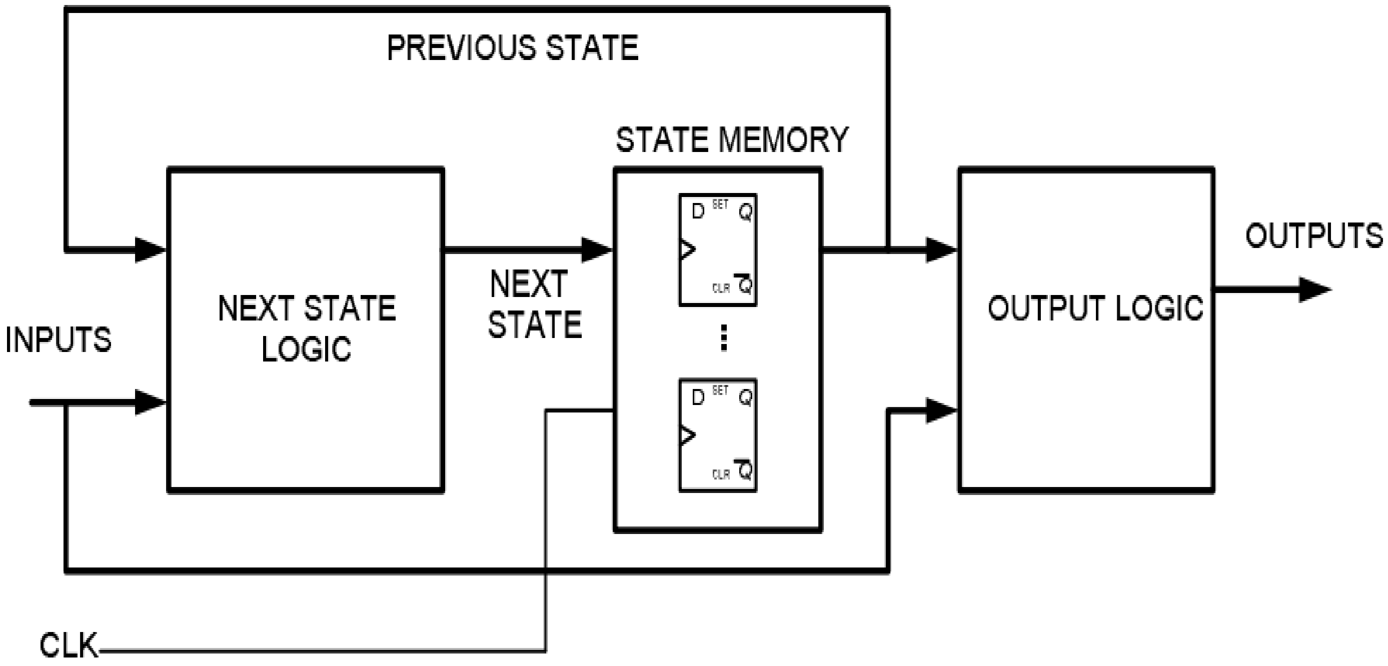
Module - 6
Design of Finite State Machine (FSM)

Module Contents

- Finite State Machine (FSM): Mealy FSM, Moore FSM
- Design Example: Sequence Detection
- Modeling of FSM using Verilog HDL.

Finite State Machine (FSM): Mealy FSM, Moore FSM

Sequential Circuit – Block Diagram

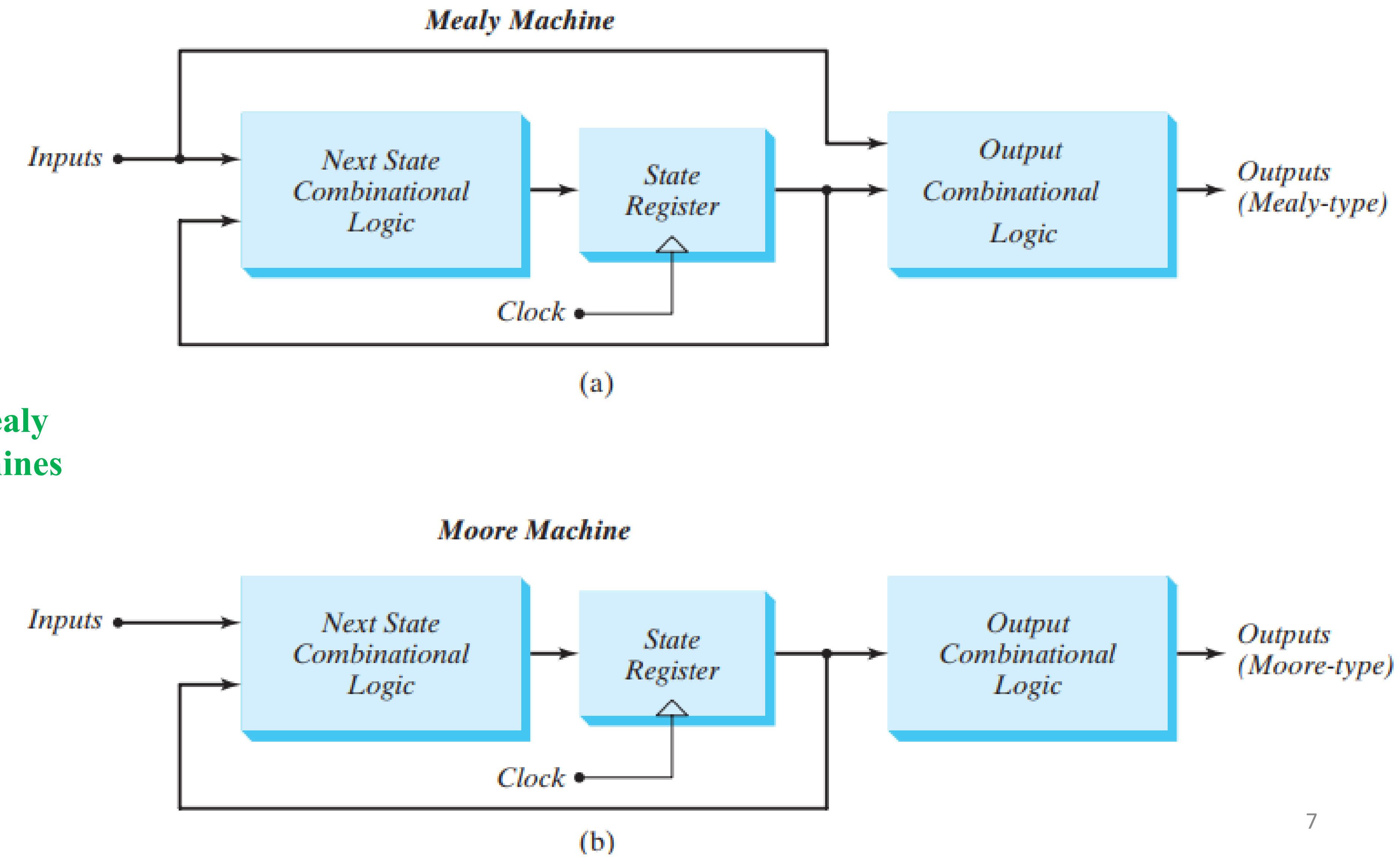


- **State memory: Set of n flip-flops that hold the state of the machine (up to 2^n distinct states)**
- **Next state logic: Combinational circuit that determines the next state as a function of the current state and the input**
- **Output logic: Combinational circuit that determines the output as a function of the current state and the input.**

Mealy and Moore Models of Finite State Machines

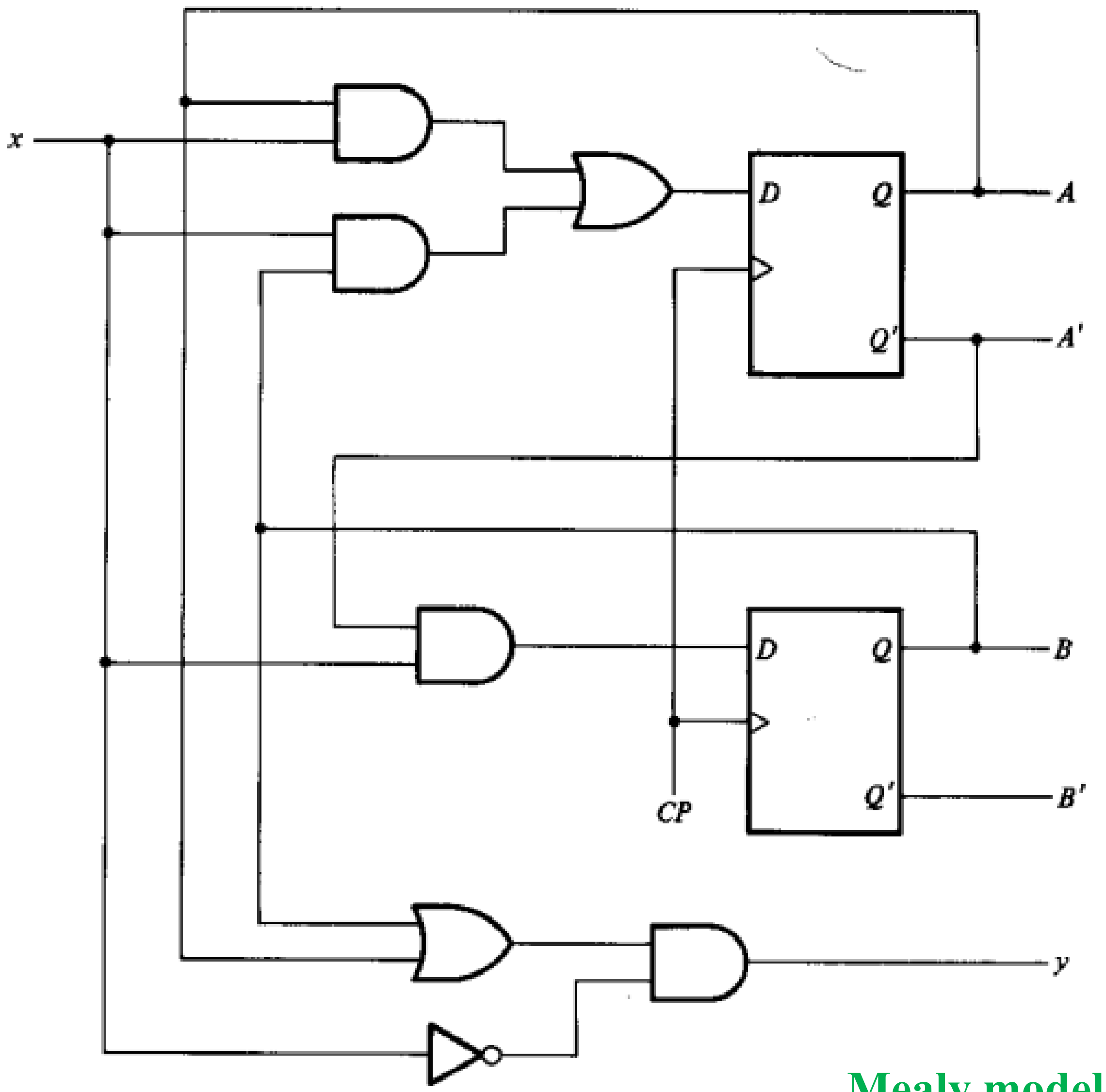
- The most general model of a sequential circuit has inputs, outputs, and internal states.
- It is customary to distinguish between two models of sequential circuits: the **Mealy model** and the **Moore model**.
- They differ only in the way the output is generated.
- In the **Mealy model**, the **output is a function of both the present state and the input**.
- In the **Moore model**, the output is a **function of only the present state**.
- A circuit may have both types of outputs. The two models of a sequential circuit are commonly referred to as a **finite state machine**, abbreviated **FSM**.
- The **Mealy model** of a sequential circuit is referred to as a **Mealy FSM or Mealy machine**. The **Moore model** is referred to as a **Moore FSM or Moore machine**.

Mealy and Moore Models of Finite State Machines

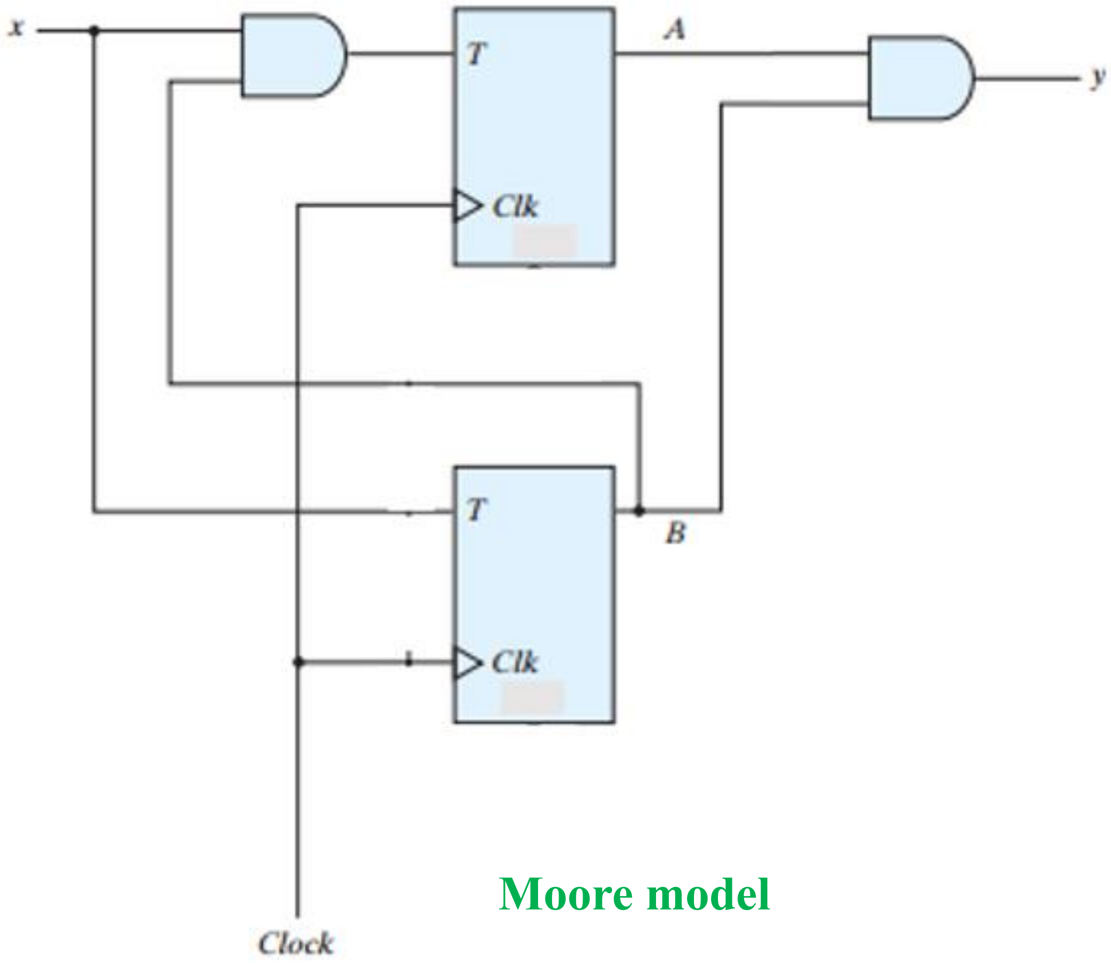


Block diagrams of (a) Mealy and (b) Moore state machines

Example:



Mealy model



Moore model

Mealy and Moore Models of Finite State Machines

- In a **Moore model**, the outputs of the sequential circuit are synchronized with the clock, because they depend only on flip-flop outputs that are synchronized with the clock.
- In a **Mealy model**, the outputs may change if the inputs change during the clock cycle. Moreover, the outputs may have momentary false values because of the delay encountered from the time that the inputs change and the time that the flip-flop outputs change.
- In order to synchronize a Mealy-type circuit, the inputs of the sequential circuit must be synchronized with the clock, and the outputs must be sampled immediately before the clock edge. The inputs are changed at the inactive edge of the clock to ensure that the inputs to the flip-flops stabilize before the active edge of the clock occurs. Thus, **the output of the Mealy machine is the value that is present immediately before the active edge of the clock.**

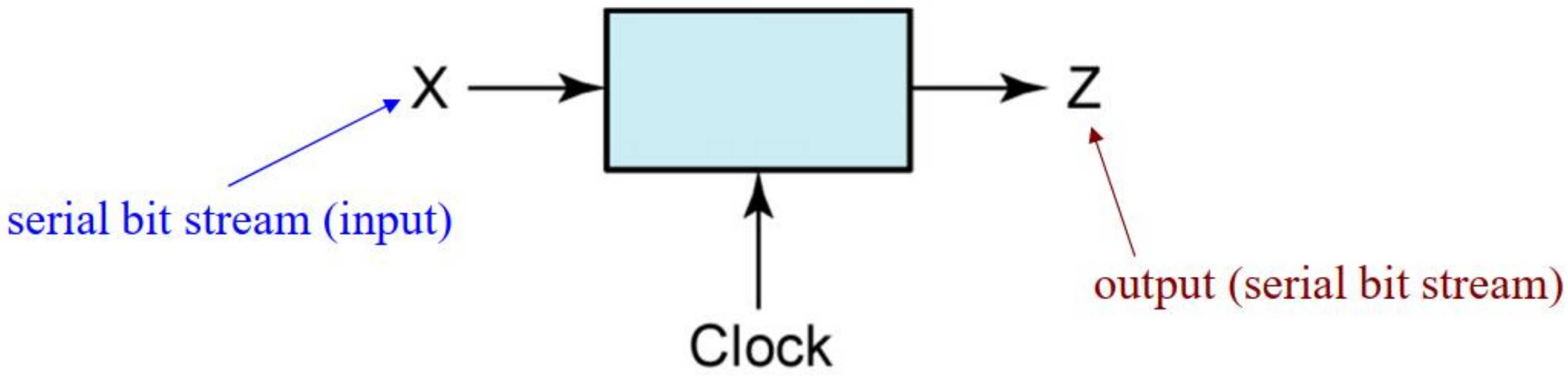
Design of Clocked Sequential Circuits

Steps:

1) From the word description and specifications of the desired operation, derive a state diagram for the circuit.
2) Reduce the number of states if necessary.
3) Assign binary values to the states.
4) Obtain the binary-coded state table.
5) Choose the type of flip-flops to be used.
6) Derive the simplified flip-flop input equations and output equations.
7) Draw the logic diagram.

Example: Sequence Detector

- To illustrate the design of a clocked Mealy sequential circuit, we will design a sequence detector.
- The circuit is of the form:



Example: A sequence detector

- Suppose we want to design the sequence detector so that any input sequence ending in **101** will produce an output of Z = 1 coincident with the last 1.
- The circuit does not reset when a 1 output occurs.
- A typical input sequence and the corresponding output sequence are:

X =	0	0	1	1	0	1	1	0	0	1	0	1	0	1	0	0
Z =	0	0	0	0	0	1	0	0	0	0	0	1	0	1	0	0
(time:	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15)

Contd…

- The sequence detector can have two cases:

  1) overlapped sequence

  2) non-overlapped sequence

- A typical input sequence and the corresponding output sequence **(overlapped)** are:

X =	0	0	1	1	0	1	1	0	0	1	0	1	0	1	0	0
Z =	0	0	0	0	0	1	0	0	0	0	0	1	0	1	0	0
(time:	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15)

Contd…

- An input sequence and the corresponding output sequence **(non-overlapped)** are:

X =	0	0	1	1	0	1	1	0	0	1	0	1	0	1	0	0
Z =	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0
(time:	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15)

Example: A sequence detector (Mealy)

- Suppose we want to design the sequence detector so that any input sequence ending in **101** will produce an output of Z = 1 coincident with the last 1.
- The circuit does not reset when a 1 output occurs.
- A typical input sequence and the corresponding output sequence are:

X =	0	0	1	1	0	1	1	0	0	1	0	1	0	1	0	0
Z =	0	0	0	0	0	1	0	0	0	0	0	1	0	1	0	0
(time:	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15)

Example: Sequence Detector (Mealy)

- **In Mealy FSM, the number of states is equal to total number of bits in the sequence.**
- Example: For "101" sequence, total number of states would be three.
- Initially, we do not know how many flip-flops will be required, so we will designate the circuit states as S0, S1, S2 etc.
- In Mealy FSM, the output depends on both state and input.

State	Input Sequence
S0	-
S1	1
S2	10

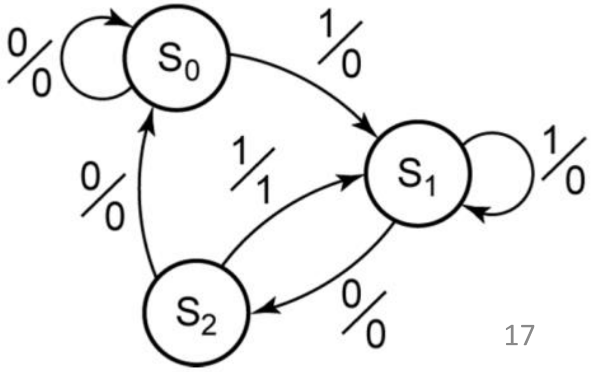
Example: Sequence Detector (Mealy)

State	Input Sequence
S0	-
S1	1
S2	10

Overlapped sequence of "101" detector

- We will start with a reset state designated **S0**.
- If a 0 input is received, the circuit can stay in S0 because the input sequence we are looking for does not start with a 0. This process produces output 0.
- If a 1 input is received, the circuit moves to S1, and this is the first bit of required sequence. This process produces output 0.
- Now we are at **S1**.
- If a 0 input is received, the circuit can stay in S2 because this gives the first two bit of required sequence. This process produces output 0.
- If a 1 input is received, the circuit moves to S1, and this is the first bit of required sequence. This process produces output 0.
- Now we are at **S2**.
- If a 0 input is received, the circuit goes to S0 because the input sequence we are looking for does not have 100. This process produces output 0.
- If a 1 input is received, the circuit moves to S1, and this is the first bit of required sequence. This process produces output 1 as the required bit is detected.

State diagram for the Mealy Machine



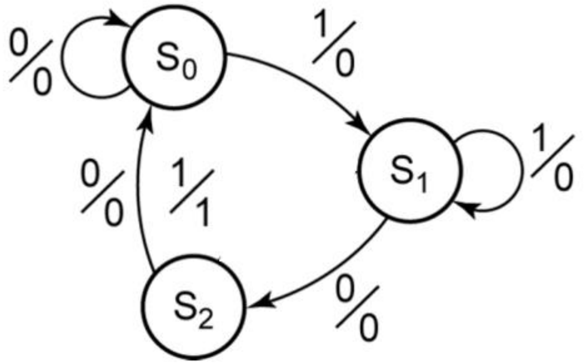
Example: Sequence Detector (Mealy)

State	Input Sequence
S0	-
S1	1
S2	10

Non-overlapped sequence of "101" detector

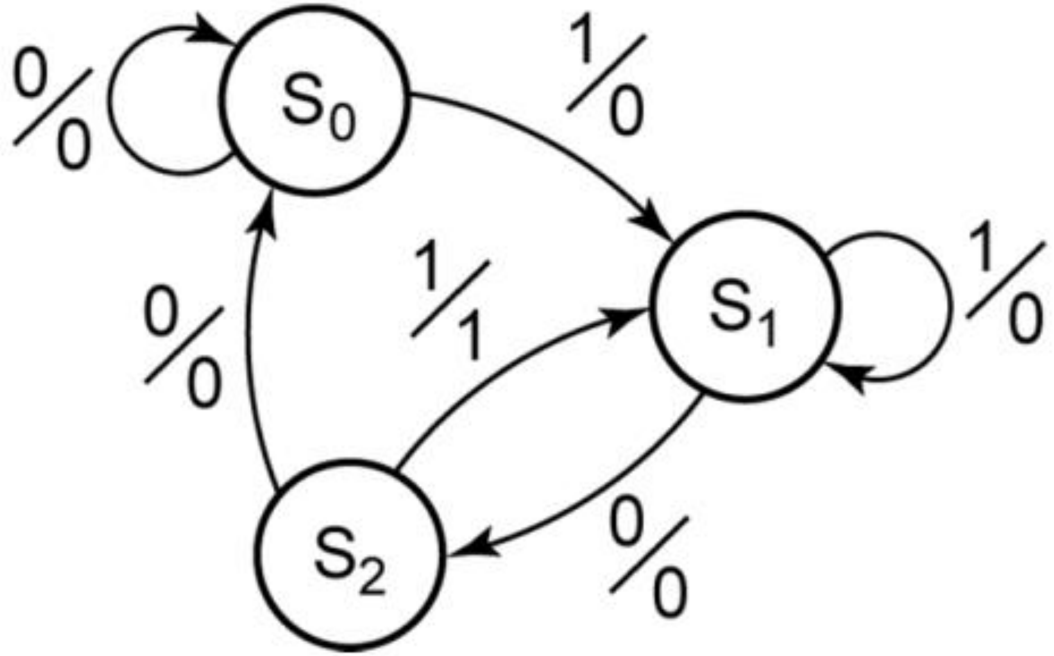
- We will start with a reset state designated **S0**.
- If a 0 input is received, the circuit can stay in S0 because the input sequence we are looking for does not start with a 0. This process produces output 0.
- If a 1 input is received, the circuit moves to S1, and this is the first bit of required sequence. This process produces output 0.
- Now we are at **S1**.
- If a 0 input is received, the circuit can stay in S2 because this gives the first two bit of required sequence. This process produces output 0.
- If a 1 input is received, the circuit moves to S1, and this is the first bit of required sequence. This process produces output 0.
- Now we are at **S2**.
- If a 0 input is received, the circuit goes to S0 because the input sequence we are looking for does not have 100. This process produces output 0.
- If a 1 input is received, the circuit moves to S0, because the "101" sequence can not be used further for another sequence, and the process should start from reset condition. This process produces output 1 as the required bit is detected.

State diagram for the Mealy Machine

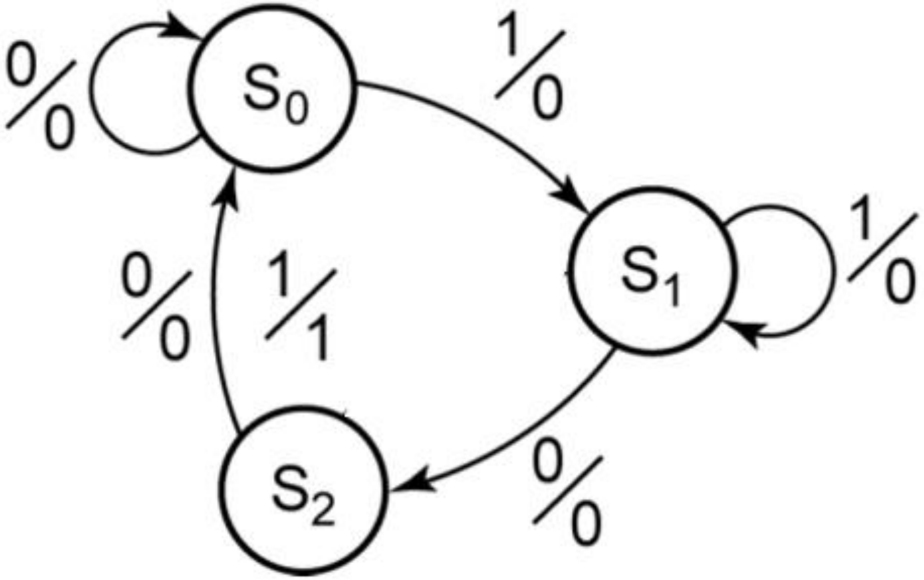


Summary (Mealy FSM)

State diagram (Overlapped 101 detector)



State diagram (Non-Overlapped 101 detector)



Example: Sequence Detector (Moore)

- **In Moore FSM, the number of states is equal to total number of bits in the sequence +1.**
- Example: For “101” sequence, total number of states would be four.
- Initially, we do not know how many flip-flops will be required, so we will designate the circuit states as S0, S1, S2, S3 etc.
- In Moore FSM, the output only depends on the state.

State	Input Sequence
S0	-
S1	1
S2	10
S3	101

Example: Sequence Detector (Moore)

Overlapped sequence of “101” detector

- **<u>We will start with a reset state designated S0. This state produces output 0.</u>**
- If a 0 input is received, the circuit can stay in S0 because the input sequence we are looking for does not start with a 0.
- If a 1 input is received, the circuit moves to S1, and this is the first bit of required sequence.
- **<u>Now we are at S1. This state produces output 0.</u>**
- If a 0 input is received, the circuit can stay in S2 because this gives the first two bit of required sequence.
- If a 1 input is received, the circuit moves to S1, and this is the first bit of required sequence.

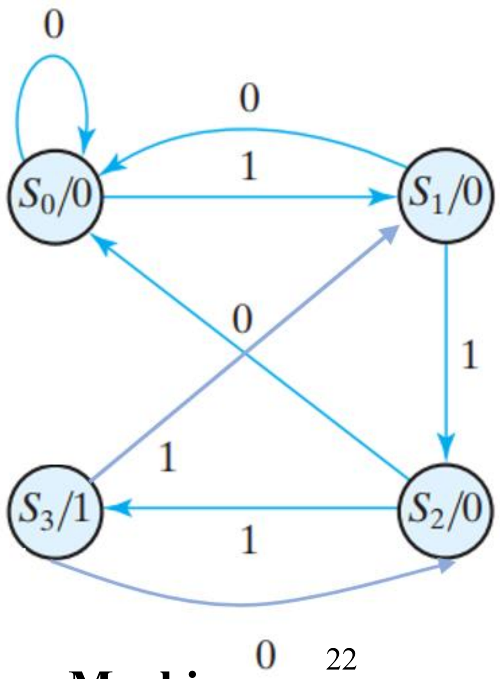
State	Input Sequence
S0	-
S1	1
S2	10
S3	101

Contd…

- **<u>Now we are at S2. This state produces output 0.</u>**
- If a 0 input is received, the circuit goes to S0 because the input sequence we are looking for does not have 100.
- If a 1 input is received, the circuit moves to S3, and this is the required sequence.
- **<u>Now we are at S3. This state produces output 1.</u>**
- If a 0 input is received, the circuit goes to S2 because the because this gives the first two bit of required sequence.
- If a 1 input is received, the circuit moves to S1, because this is the first bit of

  next required sequence.

State	Input Sequence
S0	-
S1	1
S2	10
S3	101



State diagram for the Moore Machine

Example: Sequence Detector (Moore)

Non-overlapped sequence of “101” detector

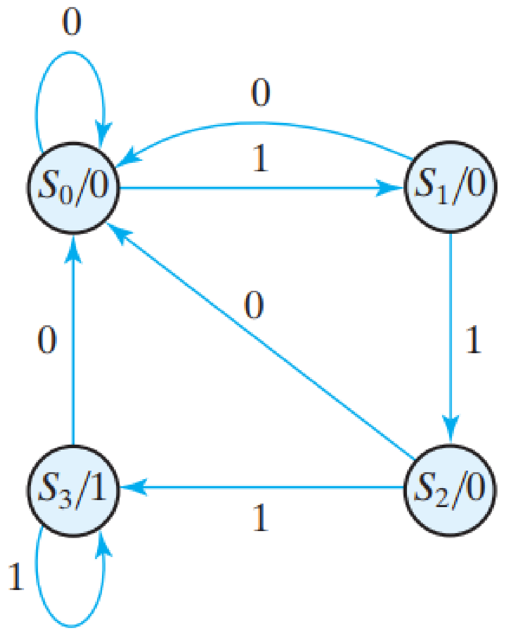
- **<u>We will start with a reset state designated S0. This state produces output 0.</u>**
- If a 0 input is received, the circuit can stay in S0 because the input sequence we are looking for does not start with a 0.
- If a 1 input is received, the circuit moves to S1, and this is the first bit of required sequence.
- **<u>Now we are at S1. This state produces output 0.</u>**
- If a 0 input is received, the circuit can stay in S2 because this gives the first two bit of required sequence.
- If a 1 input is received, the circuit moves to S1, and this is the first bit of required sequence.

State	Input Sequence
S0	-
S1	1
S2	10
S3	101

Contd…

- **<u>Now we are at S2. This state produces output 0.</u>**
- If a 0 input is received, the circuit goes to S0 because the input sequence we are looking for does not have 100.
- If a 1 input is received, the circuit moves to S3, and this is the required sequence.
- **<u>Now we are at S3. This state produces output 1.</u>**
- If a 0 input is received, the circuit goes to S0 the required sequence is already detected. We need to go to reset state for fresh search of the sequence.
- If a 1 input is received, the circuit moves to S1, because this is the first bit of next required sequence.

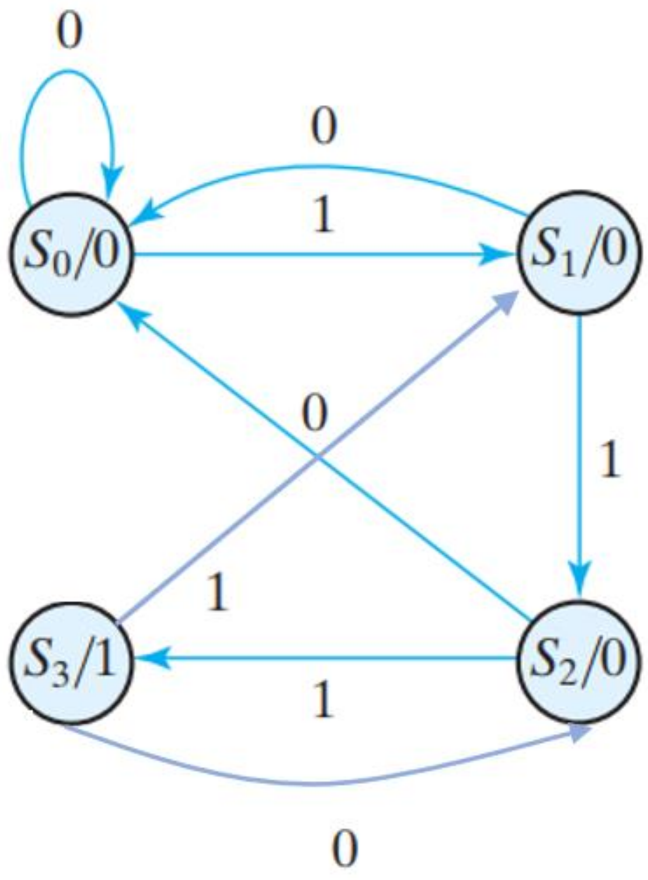
State	Input Sequence
S0	-
S1	1
S2	10
S3	101



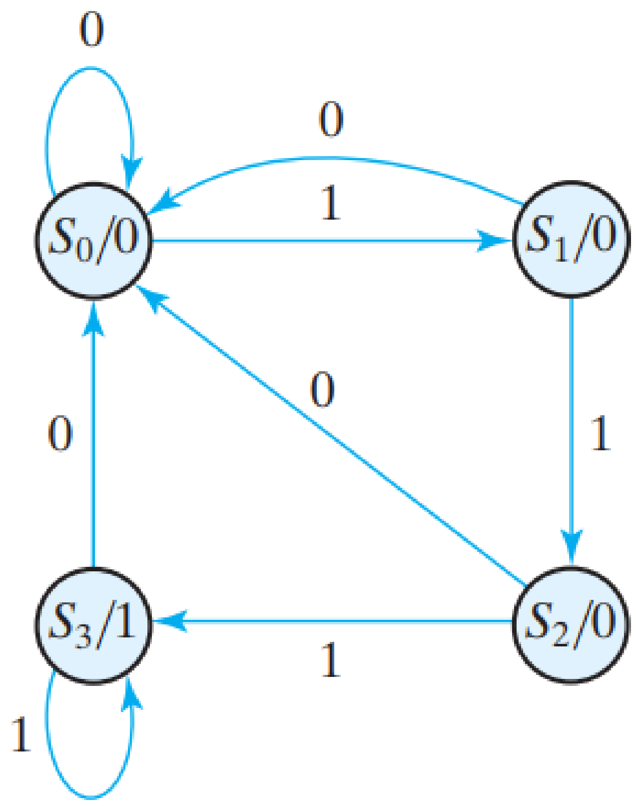
State diagram for the Moore Machine

Summary (Moore FSM)

State diagram
(Overlapped 101 detector)

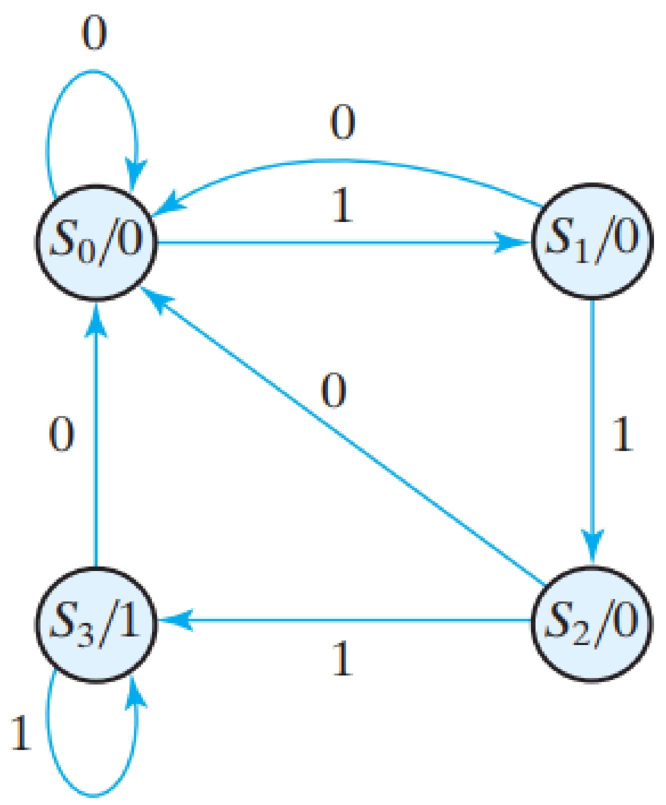


State diagram
(Non-Overlapped 101 detector)



Solution

- Let the initial state is S0, the reset state. If the input is 0, the circuit stays in S0.
- But if the input is 1, it goes to state S1 to indicate that a 1 was detected.
- If the next input is 1, the change is to state S2 to indicate the arrival of two consecutive 1's, but if the input is 0, the state goes back to S0.
- The third consecutive 1 sends the circuit to state S3. If more 1's are detected, the circuit stays in S3. Any 0 input sends the circuit back to S0.
- In this way, the circuit stays in S3 as long as there are three or more consecutive 1's received.
- This is a **Moore model sequential circuit**, since the **output is 1 when the circuit is in state S3 and is 0 otherwise**.



State diagram for sequence detector

Contd…

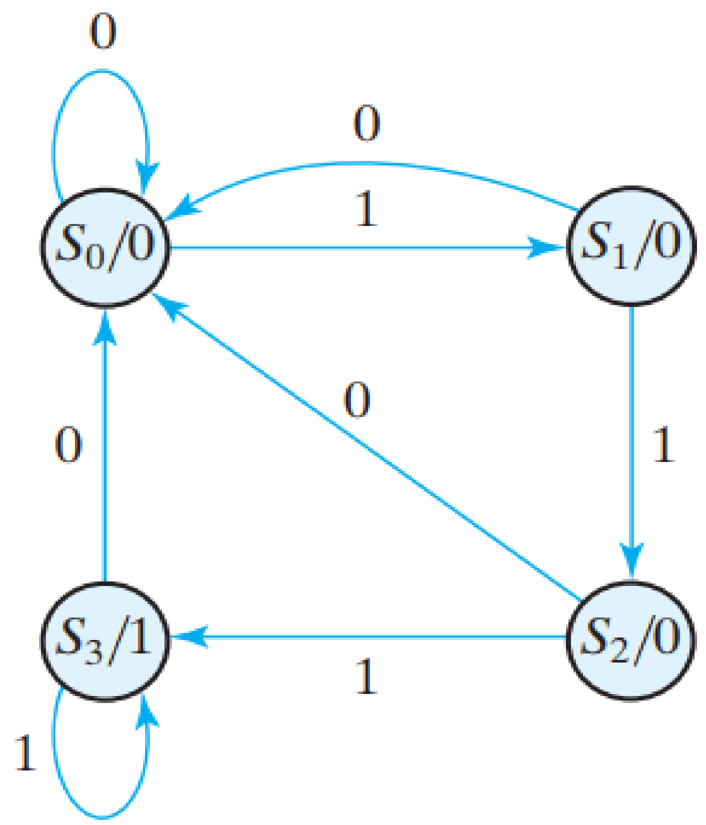
- Let the States are:

State	Binary Value
S0	00
S1	01
S2	10
S3	11

- Let the input variable is x and output variable is y

State Table for Sequence Detector

Present State		Input	Next State		Output
A	B	x	A	B	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	1	1	1	0
1	1	0	0	0	1
1	1	1	1	1	1



State diagram for sequence detector

Contd…

- Let the F/F to be used in the circuit is D F/F. As there are 4 states hence 2 F/Fs needed.
- To know the f/f input, excitation table of D-f/f is needed.

State table

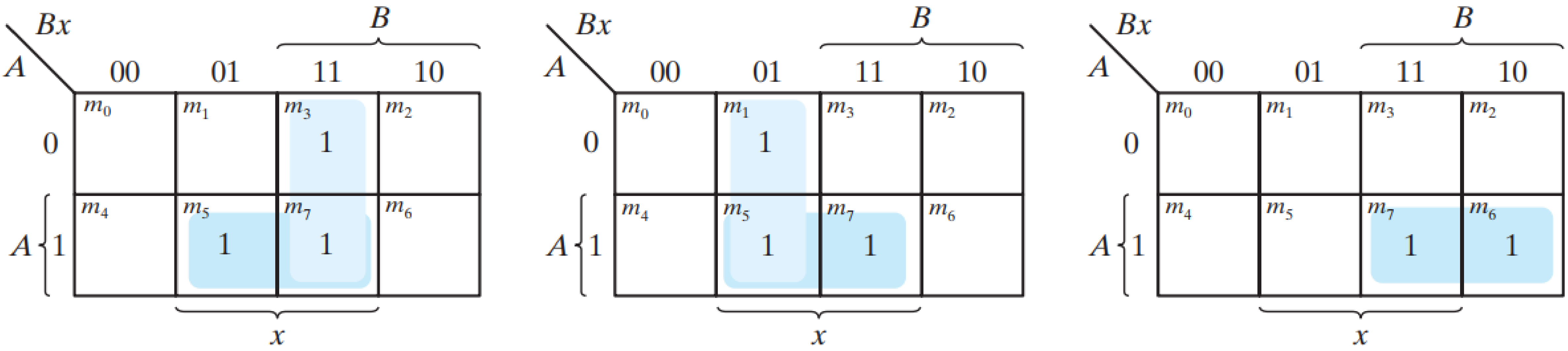
Present State		Input	Next State		Flip-Flop Input		Output
A	B	x	A	B	D_A	D_B	y
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	0
0	1	0	0	0	0	0	0
0	1	1	1	0	1	0	0
1	0	0	0	0	0	0	0
1	0	1	1	1	1	1	0
1	1	0	0	0	0	0	1
1	1	1	1	1	1	1	1

Excitation table

$Q(t)$	$Q(t+1)$	D
0	0	0
0	1	1
1	0	0
1	1	1

Contd…

- Using K-Map find the Flip-flop input equations and output equation as a function of present state and input.



$D_A = Ax + Bx$

$D_B = Ax + B'x$

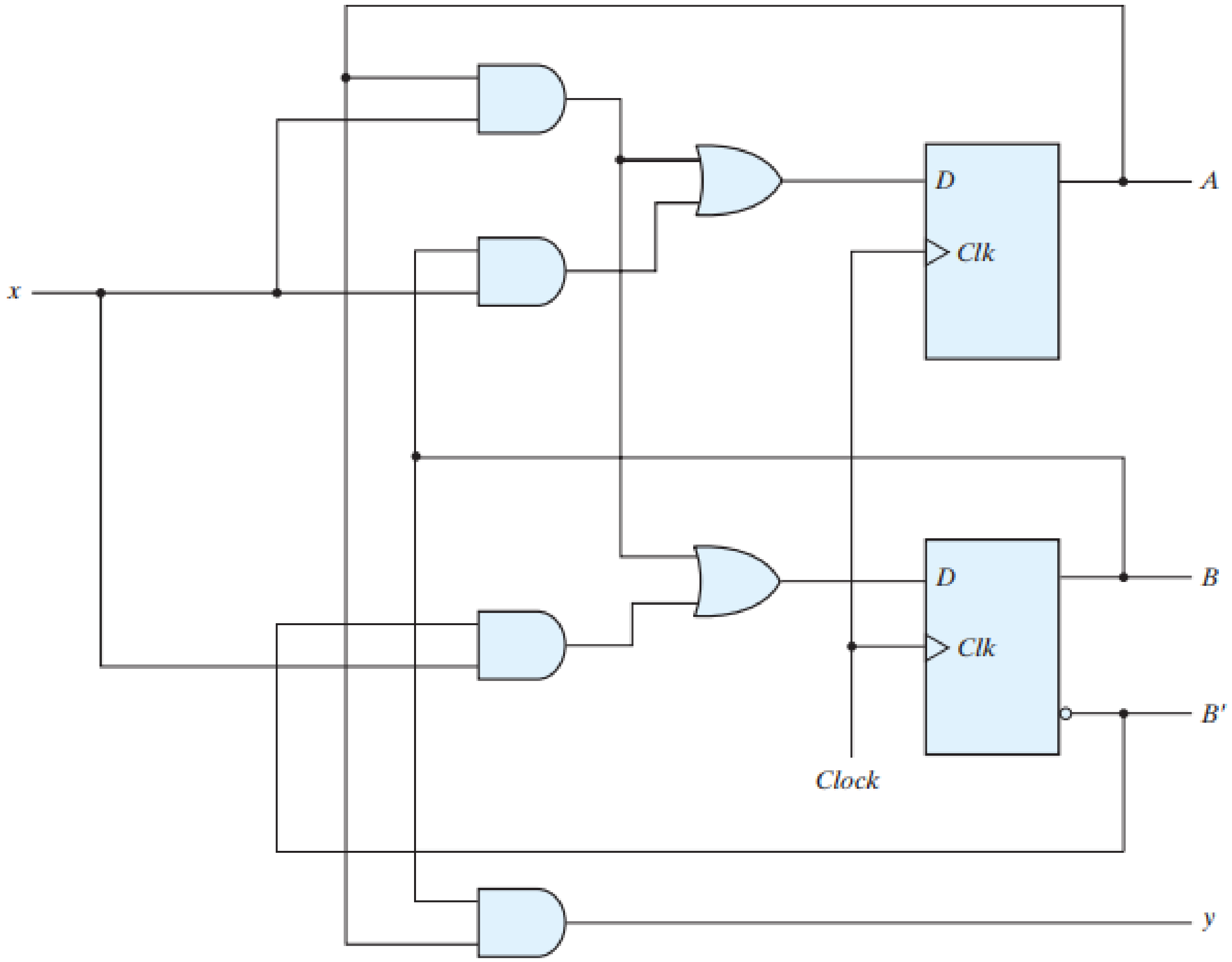
$y = AB$

Contd…

$$D_A = Ax + Bx$$
$$D_B = Ax + B'x$$
$$y = AB$$

Flip-flop input equations

Flip-flop output equation



Logic diagram of a Moore-type sequence detector

Any Queries!

Thank you!